

Stat 341 – Project Milestone 3

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```
mlvar_data = read.csv("SA_data_for_mlvar_ana.csv")
#make subjects and dyads factors
mlvar_data$subject <- as.factor(mlvar_data$subject)
mlvar_data$dyad <- as.factor(mlvar_data$dyad)
#compute synchrony, log transform to correct for heteroscedasticity
mlvar_data$Synchrony <- -log(mlvar_data$dtw)
colnames(mlvar_data) <- c('X', 'dyad', 'subject', 'dtw', 'EyeContact', 'PupilSize', 'Synchrony')

mlvar_data$Sync_standardized <- as.numeric(scale(mlvar_data$Synchrony))

mlvar_data$EC_standardized <- as.numeric(scale(mlvar_data$EyeContact))

mlvar_data$PS_standardized <- as.numeric(scale(mlvar_data$PupilSize))

mlvar_data$subjectNum <- as.numeric(mlvar_data$subject)

## we are creating a completely simulated variable. so that the project team can use a categorical pred

set.seed(50)
mlvar_data$attention <- rbinom(n = nrow(mlvar_data), size = 1, prob = 0.7) + 1

## 1: not engage, 2: engage.

glimpse(mlvar_data)

## Rows: 56,400
## Columns: 12
## $ X                <int> 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15~
## $ dyad              <fct> D001, D001, D001, D001, D001, D001, D001, D001, D001~
## $ subject           <fct> S028, S028, S028, S028, S028, S028, S028, S028, S028~
## $ dtw               <dbl> 26.038945, 42.312825, 27.318998, 37.486558, 15.10710~
## $ EyeContact        <dbl> 0.000, 0.000, 0.033, 0.256, 0.350, 0.000, 0.000, 0.0~
## $ PupilSize         <dbl> 0.35092227, 0.47939718, 0.47493465, 0.55883216, 0.37~
## $ Synchrony         <dbl> -3.2595933, -3.7450902, -3.3075824, -3.6239824, -2.7~
## $ Sync_standardized <dbl> -0.5113445, -1.0841509, -0.5679637, -0.9412636, 0.13~
## $ EC_standardized  <dbl> -0.89647023, -0.89647023, -0.80865752, -0.21525653, ~
## $ PS_standardized  <dbl> 0.98524749, 1.35187012, 1.33913560, 1.57854985, 1.05~
## $ subjectNum       <dbl> 49, 49, 49, 49, 49, 49, 49, 49, 49, 49, 49, 49, ~
## $ attention        <dbl> 1, 2, 2, 1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 1, 2~
```

The data is from a research conducted by Sophie Wohltjen and Thalia Wheatley. Their research paper can be found here: <https://www.pnas.org/doi/10.1073/pnas.2106645118>

We sourced our dataset from the author's github repository: <https://github.com/sophiewohltjen/eyeContact-in-conversation>

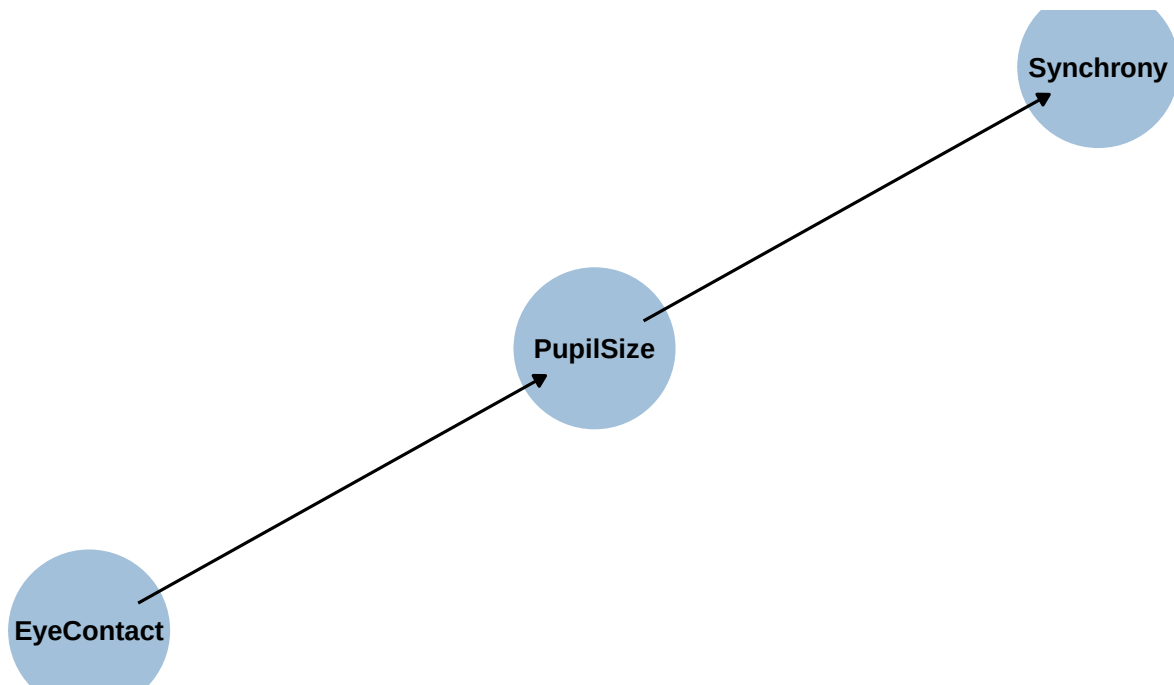
This research paper examines the role of eye contact in conversation. The authors found that when two people are talking, eye contact is a key factor in determining when shared attention between them is rising or falling. It was found that when two people are engaging in conversation, eye contact increases as the conversation progresses, indicating a rise in shared attention.

Research Question

For this project, the question we are interested in answering is: How does eye contact affect shared attention in conversations?

Our predictor variables are eye contact and pupil size the response variable is shared attention in conversation (synchrony). Our predictor variables are eye contact and pupil size. Our response variable is shared attention in conversation (synchrony).

```
precision_ex <- dagitty("dag{
  EyeContact -> PupilSize
  PupilSize -> Synchrony
}")
gg_dag(precision_ex,
  size = 30,
  highlight = 'A')
```



synchrony is used to identify attention, from the other predictors such as **PupilSize** and **EyeContact**. This diagram indicates that eye contact can influence pupil size, which in turn can influence synchronous. We decided to include **PupilSize** as a mediator.

For our predictor variables, they are already standardized, we are having trouble finding the units and range for each variable, so that would be something to think more about.

Model Description

$$\text{Synchrony}_i \sim \text{Normal}(\mu_i, \sigma)$$

$$\mu_i = \beta_0 + \beta_1 * \text{EyeContact}_i + \beta_2 * \text{PupilSize}_i + \beta_{\text{Attention}[i]}$$

$$\beta_{\text{attention}=j} \sim \text{Normal}(0, 1) \text{ for } j = 1 \dots 2$$

$$\beta_0 \sim \text{Normal}(0, 0.5)$$

$$\beta_1 \sim \text{Normal}(0, 0.5)$$

$$\beta_2 \sim \text{Normal}(0, 1)$$

$$\sigma \sim \text{lognormal}(0, 1)$$

Rationale for priors

The intercept is the expected synchrony with average eye contact and average pupil size. We expect that the average synchrony would be at 0 but with a standard deviation of 0.5 because we think we might be off by 0.5 to 1.

We put a 0.5 for our standard deviation to give some wiggle room because we aren't very certain that our data came in a perfectly normal shape and that some good amount is originally skewed.

The slope of eye contact tells that if they eye goes up by one standard deviation meaning that the duration is longer, how much will the synchrony go up? We don't have an idea whether synchrony might direct to.

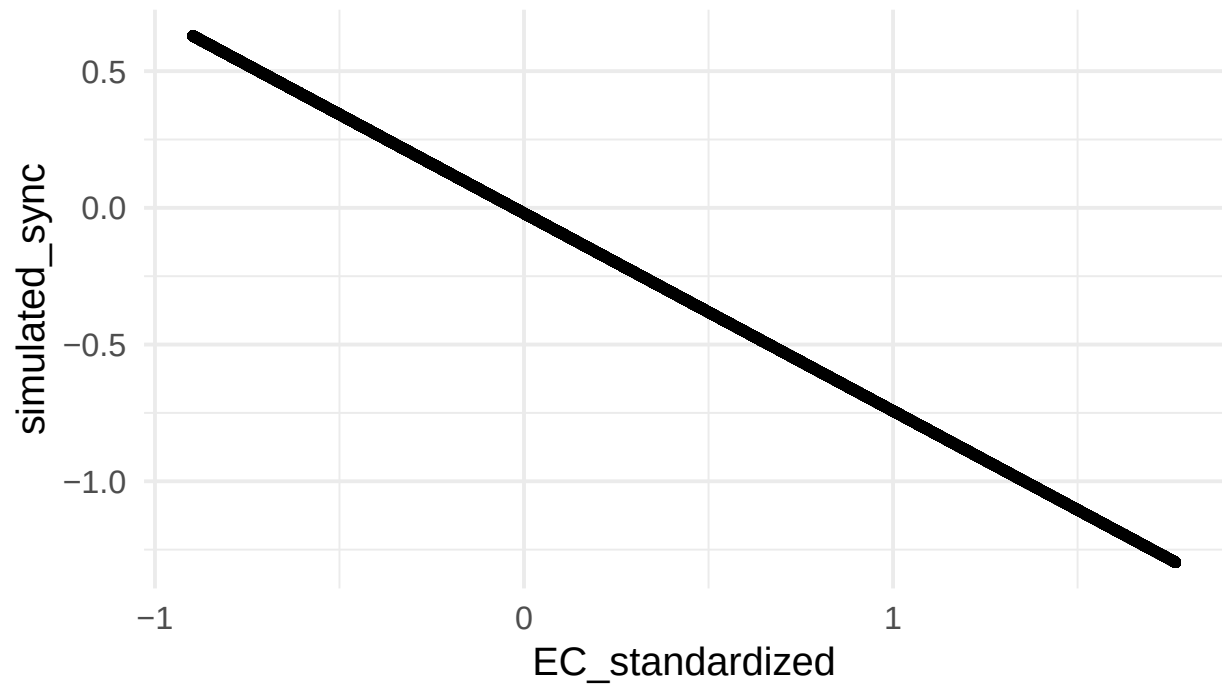
The slope for pupil size, we also weren't really sure which direction the synchrony might go based on pupil size.

Prior Predictive Check

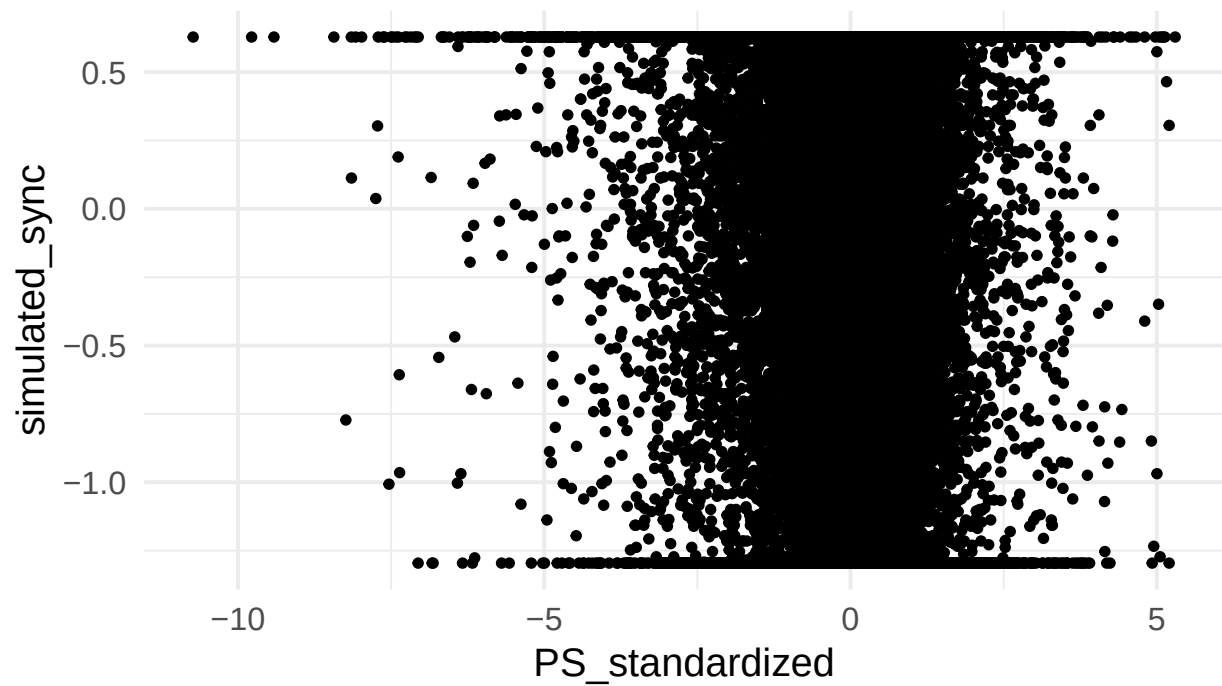
```
b0 = rnorm(1, mean = 0, sd = 0.5)
b1 = rnorm(1, mean = 0, sd = 0.5)
b2 = rnorm(1, mean = 0, sd = 1)

simulated_sync <- b0 + b1 * mlvar_data$EC_standardized + b2 * mlvar_data$EC_standardized

gf_point(simulated_sync ~ EC_standardized, data = mlvar_data)
```



```
gf_point(simulated_sync ~ PS_standardized, data = mlvar_data)
```



Interpretation

Based on our prior predictive check on eye contact, we see that with one standard deviation increasing, their synchrony increases by 0.2.

Based on the simulated prior predictive check, we see that there is a well distributed number of people in

the range and it seems like it varies in the synchrony as well despite the pupil size.

Model Fit - Stan

```
stan_data <- compose_data(mlvar_data)

stan_mlvar_program <- '
data {
  // number of observations
  int<lower=1> n;
  // response
  vector[n] Sync_standardized;
  // predictor
  vector[n] EC_standardized;
  // predictor
  vector[n] PS_standardized;
  // categorical predictor
  array[n] int attention;
}
parameters {
  // std of response, single continuous value
  real<lower=0> sigma;
  real b0;
  real b1;
  real b2;
  vector[2] b3;
}
model {
  // vector of n values: expected Synchrony for each observation
  vector[n] mu;
  // loop over the n cases in the dataset to estimate mu_i values
  for (i in 1:n) {
    mu[i] = b0 + (b1*EC_standardized[i]) + (b2*PS_standardized[i]) + b3[attention[i]];
  }

  // prior for intercept
  b0 ~ normal(0, 0.5);
  // priors for slopes
  b1 ~ normal(0, .5);
  b2 ~ normal(0, 1);
  b3 ~ normal(0, 1);
  // prior for sigma
  sigma ~ lognormal(0,1);

  // defining likelihood in terms of mus and sigma
  Sync_standardized ~ normal(mu, sigma);
}
'
```

```
mlvar_stan_model <- stan(model_code = stan_mlvar_program,
                        data = stan_data,
                        chains = 2,
```

```

iter = 500,
warmup = 250,
refresh = 0)

```

```

## Running MCMC with 2 sequential chains...
##
## Chain 1 finished in 325.4 seconds.
## Chain 2 Informational Message: The current Metropolis proposal is about to be rejected because of the
## Chain 2 Exception: normal_lpdf: Scale parameter is 0, but must be positive! (in '/tmp/Rtmp43Bro8/mod
## Chain 2 If this warning occurs sporadically, such as for highly constrained variable types like covar
## Chain 2 but if this warning occurs often then your model may be either severely ill-conditioned or m
## Chain 2
## Chain 2 Informational Message: The current Metropolis proposal is about to be rejected because of the
## Chain 2 Exception: normal_lpdf: Scale parameter is 0, but must be positive! (in '/tmp/Rtmp43Bro8/mod
## Chain 2 If this warning occurs sporadically, such as for highly constrained variable types like covar
## Chain 2 but if this warning occurs often then your model may be either severely ill-conditioned or m
## Chain 2
## Chain 2 finished in 296.5 seconds.
##
## Both chains finished successfully.
## Mean chain execution time: 311.0 seconds.
## Total execution time: 622.5 seconds.

```

```
mlvar_stan_model
```

```

## Inference for Stan model: rt_cmdstanr_b3c5e718a552bbb4ad21723705c303-202304271724-1-32ed5c.
## 2 chains, each with iter=500; warmup=250; thin=1;
## post-warmup draws per chain=250, total post-warmup draws=500.
##
##               mean se_mean   sd      2.5%      25%      50%      75%      97.5%
## sigma         0.98     0.00 0.00       0.98       0.98       0.98       0.99       0.99
## b0            -0.07     0.04 0.38      -0.75      -0.32      -0.11       0.19       0.70
## b1             0.05     0.00 0.00       0.05       0.05       0.05       0.06       0.06
## b2             0.17     0.00 0.00       0.16       0.17       0.17       0.17       0.18
## b3[1]          0.07     0.04 0.38      -0.70      -0.19       0.11       0.32       0.75
## b3[2]          0.07     0.04 0.38      -0.70      -0.19       0.11       0.32       0.75
## lp__        -27278.19     0.15 1.76 -27282.75 -27279.10 -27277.90 -27276.90 -27275.85
##               n_eff Rhat
## sigma        431 1.00
## b0           92 1.05
## b1          416 1.00
## b2          351 1.00
## b3[1]        93 1.05
## b3[2]        92 1.05
## lp__        133 1.00
##
## Samples were drawn using NUTS(diag_e) at Thu Apr 27 17:34:36 2023.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at

```

```
## convergence, Rhat=1).
```

Fit Convergence Check (Adds-on)

n_eff

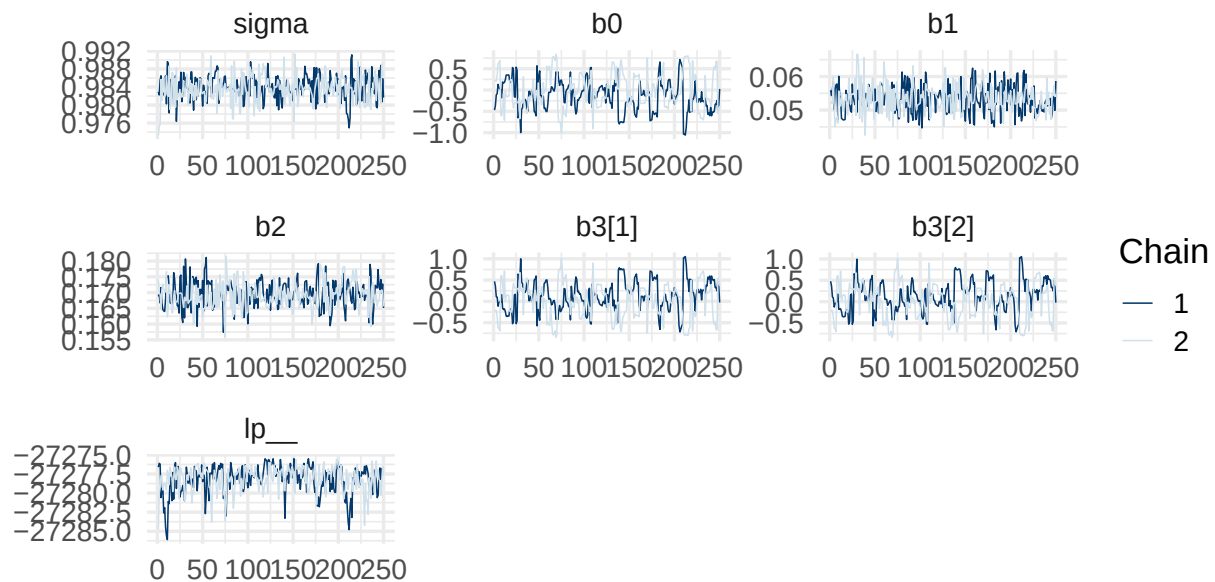
Based on the results, we see that the `n_eff` is a little bit lower than we were expecting a solution to this is to increase the number of iterations. We have a few problems with big iterations that made our code chunk to stop at the second chain because the CPU ran out of time.

R hat

For the R hat, we weren't able to remove the 1.01 due to our iterations and warmup that needed to be reduce.

Trace Plot

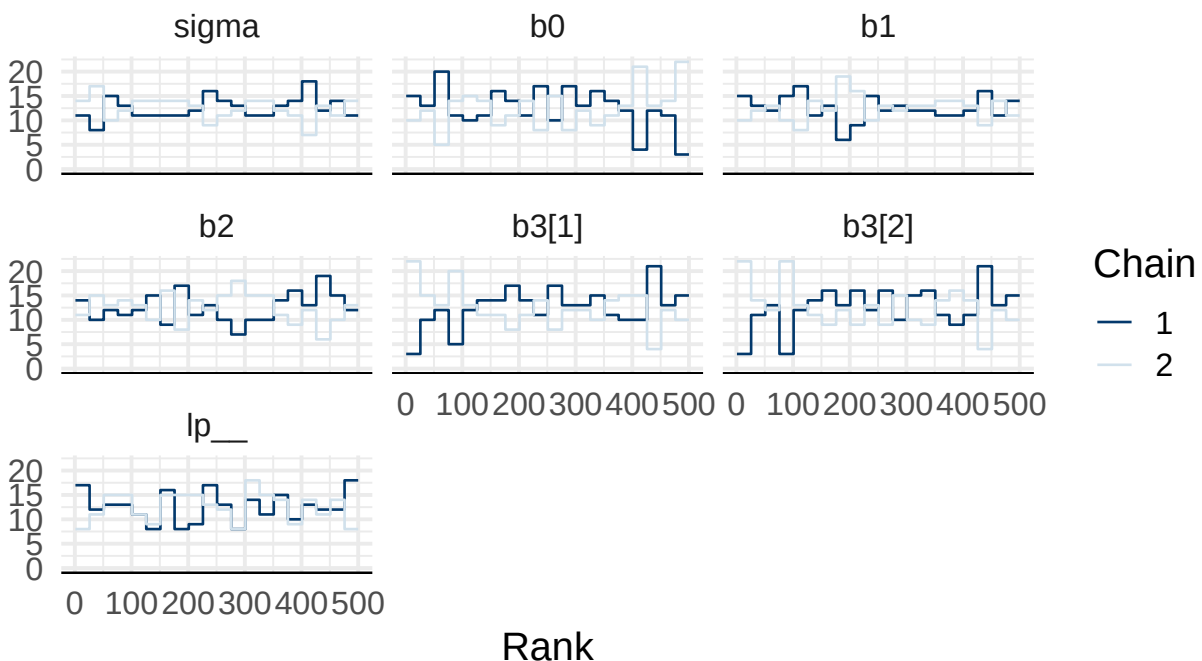
```
bayesplot::mcmc_trace(mlvar_stan_model)
```



For our trace plot, we think that both of our chain mixed well pretty well and not separate to drastically which we think is still acceptable.

Rank Trace Plot

```
bayesplot::mcmc_rank_overlay(mlvar_stan_model)
```



I think for this graph, it seems that for the intercept and the slope for the attention variable the chains are not mixed super well, I think that is something that we need to take into consideration.

Hierarchical Models (Add - Ons)

Categorical Variable and Non-Centered Stan Model

```
personLevelData <- mlvar_data |>
  group_by(subjectNum) |>
  summarise(subjectNum = first(subjectNum)) |>
  ungroup()

glimpse(personLevelData)

## Rows: 94
## Columns: 1
## $ subjectNum <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, ~

hier1_stan_data <- compose_data(
  # data at building scale (IMPORTANT to name it)
  personLevelData = personLevelData,
  # data at fine scale with response variable
  mlvar_data
)

stan_mlvar_program_CA <- '
data {
  // number of observations
  int<lower=1> n;
  int<lower=1> n_personLevelData;

  // response
```



```

vector[n] Sync_standardized;
// predictor
vector[n] EC_standardized;
// predictor
vector[n] PS_standardized;
// categorical predictor
array[n] int attention;
array[n] int subjectNum;
}
parameters {
// std of response, single continuous value
real<lower=0> sigma;
real b0;
real b1;
real b2;
real mu_person;
real<lower = 0> sigma_person;
vector[2] b3;
vector[n_personLevelData] z0;

}
model {
// vector of n values: expected Synchrony for each observation
vector[n] mu;

// loop over the n cases in the dataset to estimate mu_i value
for (i in 1:n) {
    mu[i] = b0 + mu_person + z0[subjectNum[i]] * sigma_person + b1 * EC_standardized[i] + b2 * PS_stand

}

// prior for intercept
b0 ~ normal(0, 0.5);
// priors for slopes
b1 ~ normal(0, .5);
b2 ~ normal(0,1);
b3 ~ normal(0,1);
// prior for sigma
sigma ~ lognormal(0,1);
mu_person ~ normal(0,1);
sigma_person ~ lognormal(0,1);

// defining the likelihood
z0 ~ normal(0,1);
Sync_standardized ~ normal(mu, sigma);
}
generated quantities {
    vector[n_personLevelData] a0;
    a0 = mu_person + z0 * sigma_person;
}

```

```
hier_mlvar_model <- stan(model_code = stan_mlvar_program_CA,
  data = hier1_stan_data,
  chains = 1,
  iter = 400,
  warmup = 200,
  refresh = 0)
```

```
## Running MCMC with 1 chain...
##
## Chain 1 finished in 1843.1 seconds.
```

```
hier_mlvar_model
```

```
## Inference for Stan model: rt_cmdstanr_ac47fe07e1b1a40a7ac81d93021ac7c4-202304271734-1-19bc57.
## 1 chains, each with iter=400; warmup=200; thin=1;
## post-warmup draws per chain=200, total post-warmup draws=200.
```

```
##
##               mean se_mean   sd      2.5%      25%      50%      75%
## sigma          0.87    0.00 0.00      0.87      0.87      0.87      0.87
## b0             0.04    0.03 0.42     -0.67     -0.23      0.02      0.33
## b1             0.02    0.00 0.00      0.01      0.02      0.02      0.02
## b2             0.17    0.00 0.00      0.17      0.17      0.17      0.18
## mu_person     -0.07    0.07 0.67     -1.35     -0.42     -0.12      0.33
## sigma_person   0.45    0.00 0.02      0.41      0.43      0.45      0.46
## b3[1]          0.03    0.06 0.57     -1.18     -0.31      0.02      0.41
## b3[2]          0.03    0.06 0.57     -1.19     -0.31      0.02      0.41
## z0[1]          0.28    0.05 0.14      0.02      0.18      0.27      0.37
## z0[2]          0.27    0.05 0.13      0.06      0.16      0.26      0.35
## z0[3]         -0.06    0.04 0.12     -0.32     -0.14     -0.07      0.01
## z0[4]         -0.06    0.05 0.13     -0.30     -0.14     -0.06      0.03
## z0[5]          0.89    0.05 0.13      0.65      0.81      0.89      0.99
## z0[6]          0.89    0.05 0.13      0.67      0.78      0.90      0.98
## z0[7]          0.28    0.05 0.13      0.04      0.20      0.28      0.36
## z0[8]          0.28    0.05 0.13      0.06      0.18      0.28      0.38
## z0[9]          0.09    0.05 0.13     -0.16     -0.01      0.09      0.20
## z0[10]         0.51    0.05 0.15      0.20      0.41      0.52      0.61
## z0[11]         0.12    0.05 0.14     -0.12      0.03      0.12      0.22
## z0[12]         0.51    0.05 0.13      0.29      0.41      0.52      0.62
## z0[13]         0.81    0.05 0.13      0.58      0.72      0.81      0.91
## z0[14]         0.80    0.05 0.13      0.54      0.70      0.80      0.89
## z0[15]        -0.53    0.05 0.13     -0.77     -0.63     -0.53     -0.44
## z0[16]        -0.53    0.04 0.13     -0.77     -0.62     -0.53     -0.45
## z0[17]        -1.89    0.05 0.16     -2.22     -2.01     -1.89     -1.78
## z0[18]        -1.90    0.05 0.15     -2.16     -2.01     -1.91     -1.79
## z0[19]        -1.68    0.05 0.15     -1.97     -1.78     -1.67     -1.58
## z0[20]        -1.68    0.05 0.15     -1.99     -1.78     -1.67     -1.57
## z0[21]         0.09    0.05 0.13     -0.17     -0.02      0.08      0.18
## z0[22]         0.52    0.05 0.13      0.28      0.43      0.52      0.60
## z0[23]         0.09    0.05 0.13     -0.14     -0.01      0.09      0.19
## z0[24]         0.51    0.05 0.14      0.26      0.42      0.52      0.63
## z0[25]         0.02    0.05 0.13     -0.20     -0.09      0.02      0.12
## z0[26]         0.02    0.05 0.14     -0.23     -0.07      0.01      0.11
## z0[27]        -1.58    0.05 0.15     -1.89     -1.67     -1.59     -1.48
## z0[28]        -1.54    0.05 0.15     -1.81     -1.65     -1.54     -1.44
## z0[29]        -1.58    0.05 0.15     -1.85     -1.69     -1.58     -1.48
```

## z0[30]	-1.54	0.05	0.15	-1.80	-1.64	-1.54	-1.43
## z0[31]	-0.51	0.05	0.13	-0.75	-0.61	-0.50	-0.41
## z0[32]	2.19	0.06	0.17	1.87	2.08	2.19	2.30
## z0[33]	-2.19	0.05	0.16	-2.49	-2.29	-2.19	-2.08
## z0[34]	-0.50	0.05	0.13	-0.74	-0.62	-0.50	-0.41
## z0[35]	-2.18	0.05	0.16	-2.48	-2.27	-2.18	-2.07
## z0[36]	0.96	0.05	0.13	0.72	0.86	0.97	1.05
## z0[37]	2.19	0.05	0.16	1.89	2.09	2.20	2.29
## z0[38]	-0.61	0.05	0.13	-0.84	-0.72	-0.60	-0.51
## z0[39]	0.96	0.05	0.13	0.73	0.87	0.95	1.05
## z0[40]	-1.16	0.05	0.14	-1.44	-1.25	-1.16	-1.07
## z0[41]	-0.62	0.05	0.13	-0.87	-0.71	-0.61	-0.53
## z0[42]	-1.15	0.04	0.14	-1.43	-1.24	-1.14	-1.05
## z0[43]	0.08	0.05	0.13	-0.16	-0.02	0.05	0.20
## z0[44]	0.08	0.05	0.13	-0.18	-0.01	0.07	0.17
## z0[45]	-0.89	0.05	0.13	-1.11	-0.97	-0.89	-0.81
## z0[46]	-0.88	0.05	0.13	-1.16	-0.96	-0.88	-0.80
## z0[47]	0.65	0.05	0.14	0.37	0.55	0.67	0.75
## z0[48]	0.66	0.05	0.14	0.41	0.56	0.65	0.76
## z0[49]	-0.08	0.05	0.12	-0.30	-0.18	-0.09	0.03
## z0[50]	-0.07	0.05	0.13	-0.30	-0.16	-0.08	0.02
## z0[51]	-0.47	0.05	0.13	-0.74	-0.56	-0.48	-0.38
## z0[52]	-0.49	0.05	0.13	-0.72	-0.58	-0.49	-0.41
## z0[53]	-0.47	0.05	0.13	-0.71	-0.55	-0.47	-0.36
## z0[54]	-0.50	0.04	0.12	-0.77	-0.59	-0.49	-0.42
## z0[55]	-1.83	0.05	0.15	-2.13	-1.94	-1.83	-1.73
## z0[56]	-1.83	0.05	0.15	-2.10	-1.95	-1.83	-1.71
## z0[57]	-0.19	0.05	0.13	-0.43	-0.29	-0.18	-0.09
## z0[58]	-0.20	0.04	0.12	-0.44	-0.28	-0.20	-0.12
## z0[59]	-0.64	0.05	0.13	-0.86	-0.73	-0.65	-0.55
## z0[60]	-0.52	0.05	0.13	-0.78	-0.61	-0.53	-0.43
## z0[61]	1.00	0.06	0.14	0.77	0.90	0.98	1.10
## z0[62]	1.00	0.05	0.13	0.76	0.92	0.99	1.09
## z0[63]	0.12	0.05	0.13	-0.10	0.03	0.12	0.21
## z0[64]	0.12	0.05	0.13	-0.11	0.04	0.12	0.21
## z0[65]	1.78	0.06	0.15	1.46	1.67	1.78	1.87
## z0[66]	1.78	0.05	0.15	1.47	1.69	1.79	1.87
## z0[67]	0.75	0.05	0.14	0.49	0.65	0.75	0.83
## z0[68]	0.74	0.05	0.14	0.49	0.65	0.73	0.84
## z0[69]	0.06	0.05	0.13	-0.18	-0.04	0.05	0.16
## z0[70]	0.03	0.05	0.14	-0.24	-0.07	0.03	0.14
## z0[71]	0.68	0.06	0.14	0.44	0.58	0.66	0.79
## z0[72]	0.69	0.05	0.13	0.42	0.61	0.68	0.78
## z0[73]	0.35	0.05	0.15	0.08	0.23	0.34	0.47
## z0[74]	0.33	0.05	0.13	0.09	0.23	0.34	0.44
## z0[75]	0.37	0.05	0.13	0.15	0.27	0.37	0.46
## z0[76]	0.37	0.05	0.13	0.14	0.27	0.37	0.46
## z0[77]	0.48	0.05	0.14	0.25	0.37	0.48	0.58
## z0[78]	0.47	0.05	0.13	0.25	0.36	0.46	0.58
## z0[79]	2.11	0.06	0.17	1.84	1.98	2.12	2.21
## z0[80]	2.10	0.06	0.15	1.83	2.00	2.10	2.19
## z0[81]	0.18	0.05	0.14	-0.07	0.08	0.18	0.29
## z0[82]	0.17	0.05	0.13	-0.07	0.07	0.17	0.25
## z0[83]	-0.16	0.05	0.13	-0.42	-0.26	-0.15	-0.05

## z0[84]	-0.17	0.05	0.13	-0.40	-0.26	-0.17	-0.07
## z0[85]	0.60	0.05	0.13	0.36	0.50	0.60	0.72
## z0[86]	0.60	0.05	0.12	0.40	0.52	0.58	0.69
## z0[87]	-0.41	0.05	0.13	-0.69	-0.51	-0.41	-0.31
## z0[88]	-0.42	0.05	0.13	-0.69	-0.52	-0.41	-0.32
## z0[89]	1.89	0.05	0.15	1.62	1.80	1.88	1.98
## z0[90]	1.88	0.05	0.15	1.60	1.78	1.88	1.97
## z0[91]	0.81	0.05	0.13	0.58	0.71	0.80	0.91
## z0[92]	0.84	0.05	0.13	0.62	0.74	0.83	0.93
## z0[93]	-1.71	0.05	0.15	-2.03	-1.81	-1.69	-1.60
## z0[94]	-1.72	0.05	0.15	-2.00	-1.83	-1.73	-1.61
## a0[1]	0.06	0.07	0.67	-1.24	-0.30	0.00	0.45
## a0[2]	0.05	0.07	0.67	-1.24	-0.33	0.00	0.42
## a0[3]	-0.09	0.07	0.68	-1.40	-0.46	-0.15	0.28
## a0[4]	-0.09	0.07	0.67	-1.39	-0.45	-0.15	0.29
## a0[5]	0.33	0.07	0.67	-0.98	-0.04	0.28	0.70
## a0[6]	0.33	0.07	0.67	-0.97	-0.03	0.27	0.70
## a0[7]	0.06	0.07	0.67	-1.33	-0.32	0.01	0.47
## a0[8]	0.06	0.07	0.67	-1.26	-0.33	0.02	0.47
## a0[9]	-0.03	0.07	0.67	-1.31	-0.37	-0.09	0.39
## a0[10]	0.16	0.07	0.67	-1.11	-0.22	0.12	0.54
## a0[11]	-0.01	0.07	0.67	-1.29	-0.40	-0.07	0.38
## a0[12]	0.16	0.07	0.67	-1.13	-0.23	0.09	0.57
## a0[13]	0.30	0.07	0.67	-1.03	-0.07	0.24	0.71
## a0[14]	0.29	0.07	0.67	-1.00	-0.08	0.23	0.68
## a0[15]	-0.30	0.07	0.67	-1.62	-0.64	-0.34	0.07
## a0[16]	-0.30	0.07	0.67	-1.64	-0.64	-0.36	0.08
## a0[17]	-0.91	0.07	0.68	-2.24	-1.30	-0.95	-0.54
## a0[18]	-0.92	0.07	0.67	-2.20	-1.28	-0.98	-0.53
## a0[19]	-0.82	0.07	0.67	-2.12	-1.19	-0.87	-0.45
## a0[20]	-0.82	0.07	0.67	-2.13	-1.19	-0.88	-0.44
## a0[21]	-0.03	0.07	0.67	-1.35	-0.41	-0.08	0.37
## a0[22]	0.17	0.07	0.67	-1.18	-0.19	0.10	0.55
## a0[23]	-0.03	0.07	0.67	-1.31	-0.40	-0.10	0.36
## a0[24]	0.16	0.07	0.67	-1.13	-0.21	0.10	0.54
## a0[25]	-0.06	0.07	0.67	-1.42	-0.41	-0.11	0.32
## a0[26]	-0.06	0.07	0.67	-1.41	-0.45	-0.11	0.34
## a0[27]	-0.77	0.07	0.67	-2.08	-1.14	-0.82	-0.36
## a0[28]	-0.76	0.07	0.67	-2.07	-1.15	-0.80	-0.37
## a0[29]	-0.77	0.07	0.67	-2.09	-1.16	-0.83	-0.39
## a0[30]	-0.75	0.07	0.67	-2.04	-1.12	-0.82	-0.36
## a0[31]	-0.29	0.07	0.67	-1.59	-0.67	-0.34	0.10
## a0[32]	0.91	0.07	0.67	-0.33	0.54	0.86	1.31
## a0[33]	-1.05	0.07	0.67	-2.39	-1.43	-1.10	-0.66
## a0[34]	-0.29	0.07	0.67	-1.63	-0.65	-0.33	0.09
## a0[35]	-1.04	0.07	0.67	-2.35	-1.41	-1.11	-0.64
## a0[36]	0.36	0.07	0.67	-0.99	-0.04	0.31	0.77
## a0[37]	0.91	0.07	0.67	-0.37	0.55	0.86	1.31
## a0[38]	-0.34	0.07	0.67	-1.61	-0.68	-0.39	0.03
## a0[39]	0.36	0.07	0.67	-0.93	0.01	0.31	0.76
## a0[40]	-0.58	0.07	0.67	-1.82	-0.96	-0.64	-0.22
## a0[41]	-0.34	0.07	0.67	-1.69	-0.73	-0.39	0.05
## a0[42]	-0.58	0.07	0.67	-1.95	-0.92	-0.64	-0.18
## a0[43]	-0.03	0.07	0.67	-1.35	-0.39	-0.09	0.38

## a0[44]	-0.03	0.07	0.67	-1.32	-0.41	-0.10	0.33
## a0[45]	-0.46	0.07	0.67	-1.76	-0.82	-0.52	-0.05
## a0[46]	-0.46	0.07	0.67	-1.78	-0.85	-0.52	-0.07
## a0[47]	0.22	0.07	0.67	-1.06	-0.16	0.17	0.61
## a0[48]	0.23	0.07	0.67	-1.09	-0.13	0.17	0.60
## a0[49]	-0.10	0.07	0.67	-1.43	-0.47	-0.16	0.31
## a0[50]	-0.10	0.07	0.67	-1.43	-0.46	-0.15	0.29
## a0[51]	-0.28	0.07	0.67	-1.55	-0.65	-0.33	0.09
## a0[52]	-0.29	0.07	0.67	-1.60	-0.67	-0.34	0.11
## a0[53]	-0.28	0.07	0.67	-1.55	-0.64	-0.32	0.09
## a0[54]	-0.29	0.07	0.67	-1.60	-0.67	-0.36	0.11
## a0[55]	-0.88	0.07	0.67	-2.16	-1.23	-0.94	-0.51
## a0[56]	-0.88	0.07	0.67	-2.14	-1.26	-0.94	-0.51
## a0[57]	-0.15	0.07	0.67	-1.49	-0.52	-0.20	0.22
## a0[58]	-0.16	0.07	0.67	-1.47	-0.53	-0.22	0.22
## a0[59]	-0.35	0.07	0.67	-1.69	-0.72	-0.41	0.05
## a0[60]	-0.30	0.07	0.67	-1.58	-0.68	-0.37	0.11
## a0[61]	0.38	0.07	0.67	-0.90	0.00	0.32	0.77
## a0[62]	0.38	0.07	0.67	-0.90	0.02	0.32	0.77
## a0[63]	-0.01	0.07	0.67	-1.33	-0.38	-0.08	0.37
## a0[64]	-0.01	0.07	0.66	-1.31	-0.41	-0.07	0.38
## a0[65]	0.73	0.07	0.67	-0.60	0.37	0.68	1.11
## a0[66]	0.73	0.07	0.67	-0.57	0.36	0.66	1.13
## a0[67]	0.27	0.07	0.67	-1.06	-0.10	0.21	0.66
## a0[68]	0.27	0.07	0.67	-1.08	-0.10	0.23	0.66
## a0[69]	-0.04	0.07	0.67	-1.30	-0.41	-0.10	0.34
## a0[70]	-0.05	0.07	0.67	-1.35	-0.41	-0.10	0.34
## a0[71]	0.24	0.07	0.67	-1.10	-0.13	0.19	0.61
## a0[72]	0.24	0.07	0.67	-1.05	-0.12	0.18	0.61
## a0[73]	0.09	0.07	0.67	-1.20	-0.28	0.03	0.49
## a0[74]	0.08	0.07	0.67	-1.23	-0.28	0.02	0.49
## a0[75]	0.10	0.07	0.67	-1.23	-0.27	0.06	0.49
## a0[76]	0.10	0.07	0.67	-1.17	-0.28	0.05	0.49
## a0[77]	0.15	0.07	0.67	-1.14	-0.22	0.10	0.54
## a0[78]	0.14	0.07	0.67	-1.18	-0.22	0.07	0.55
## a0[79]	0.88	0.07	0.67	-0.37	0.52	0.81	1.28
## a0[80]	0.87	0.07	0.67	-0.49	0.52	0.81	1.26
## a0[81]	0.02	0.07	0.67	-1.32	-0.37	-0.04	0.41
## a0[82]	0.01	0.07	0.67	-1.27	-0.37	-0.04	0.39
## a0[83]	-0.14	0.07	0.67	-1.41	-0.49	-0.20	0.27
## a0[84]	-0.14	0.07	0.67	-1.42	-0.50	-0.20	0.23
## a0[85]	0.20	0.07	0.67	-1.13	-0.15	0.17	0.58
## a0[86]	0.20	0.07	0.67	-1.17	-0.16	0.14	0.58
## a0[87]	-0.25	0.07	0.67	-1.55	-0.63	-0.30	0.12
## a0[88]	-0.26	0.07	0.67	-1.54	-0.61	-0.31	0.14
## a0[89]	0.78	0.07	0.67	-0.57	0.41	0.71	1.15
## a0[90]	0.78	0.07	0.67	-0.50	0.40	0.72	1.17
## a0[91]	0.30	0.07	0.67	-1.05	-0.08	0.23	0.68
## a0[92]	0.31	0.07	0.67	-0.96	-0.06	0.25	0.70
## a0[93]	-0.83	0.07	0.67	-2.14	-1.20	-0.89	-0.44
## a0[94]	-0.83	0.07	0.67	-2.16	-1.21	-0.88	-0.46
## lp_	-20438.67	1.25	7.83	-20452.40	-20444.35	-20439.40	-20432.80
##	97.5% n_eff Rhat						
## sigma	0.88	232	1.00				

## b0	1.03	161	1.01
## b1	0.03	190	0.99
## b2	0.18	184	1.00
## mu_person	1.21	90	1.00
## sigma_person	0.49	18	1.16
## b3[1]	1.08	77	1.03
## b3[2]	1.10	77	1.03
## z0[1]	0.56	8	1.00
## z0[2]	0.54	8	1.00
## z0[3]	0.21	10	1.01
## z0[4]	0.16	7	1.00
## z0[5]	1.14	7	1.00
## z0[6]	1.11	6	1.00
## z0[7]	0.52	7	1.00
## z0[8]	0.51	7	1.00
## z0[9]	0.32	7	1.00
## z0[10]	0.81	8	0.99
## z0[11]	0.39	7	1.00
## z0[12]	0.75	6	1.00
## z0[13]	1.07	7	1.00
## z0[14]	1.05	7	1.00
## z0[15]	-0.30	7	1.02
## z0[16]	-0.28	8	1.01
## z0[17]	-1.59	11	1.08
## z0[18]	-1.58	10	1.09
## z0[19]	-1.41	9	1.07
## z0[20]	-1.41	10	1.05
## z0[21]	0.33	7	1.00
## z0[22]	0.76	7	1.00
## z0[23]	0.31	7	1.00
## z0[24]	0.74	7	1.00
## z0[25]	0.24	7	1.00
## z0[26]	0.29	8	1.01
## z0[27]	-1.27	9	1.04
## z0[28]	-1.26	9	1.04
## z0[29]	-1.30	9	1.06
## z0[30]	-1.26	9	1.06
## z0[31]	-0.26	8	1.00
## z0[32]	2.53	8	1.03
## z0[33]	-1.88	12	1.08
## z0[34]	-0.25	8	1.00
## z0[35]	-1.87	11	1.07
## z0[36]	1.19	6	1.00
## z0[37]	2.50	9	1.04
## z0[38]	-0.37	8	1.01
## z0[39]	1.20	6	1.01
## z0[40]	-0.90	10	1.02
## z0[41]	-0.38	8	1.03
## z0[42]	-0.90	10	1.05
## z0[43]	0.32	7	1.00
## z0[44]	0.32	7	1.00
## z0[45]	-0.62	8	1.04
## z0[46]	-0.64	8	1.02
## z0[47]	0.87	7	1.00

## z0[48]	0.94	7 1.00
## z0[49]	0.15	6 1.00
## z0[50]	0.17	7 1.00
## z0[51]	-0.22	8 1.01
## z0[52]	-0.24	7 1.02
## z0[53]	-0.25	8 1.01
## z0[54]	-0.25	8 1.02
## z0[55]	-1.54	10 1.03
## z0[56]	-1.54	10 1.12
## z0[57]	0.02	7 1.00
## z0[58]	0.04	8 1.02
## z0[59]	-0.38	7 1.02
## z0[60]	-0.28	8 1.01
## z0[61]	1.29	6 1.00
## z0[62]	1.24	7 1.00
## z0[63]	0.37	6 1.00
## z0[64]	0.36	7 1.00
## z0[65]	2.11	8 1.02
## z0[66]	2.11	8 1.03
## z0[67]	1.03	8 1.00
## z0[68]	1.03	8 1.00
## z0[69]	0.31	7 1.00
## z0[70]	0.28	7 1.00
## z0[71]	0.95	6 1.00
## z0[72]	0.94	7 1.00
## z0[73]	0.62	7 1.00
## z0[74]	0.56	6 1.00
## z0[75]	0.64	7 1.00
## z0[76]	0.60	7 1.00
## z0[77]	0.72	7 1.00
## z0[78]	0.70	7 1.00
## z0[79]	2.46	8 1.03
## z0[80]	2.43	7 1.05
## z0[81]	0.42	7 1.00
## z0[82]	0.40	7 1.00
## z0[83]	0.07	7 1.00
## z0[84]	0.10	8 1.00
## z0[85]	0.81	7 1.00
## z0[86]	0.85	6 1.00
## z0[87]	-0.19	7 1.01
## z0[88]	-0.19	7 1.00
## z0[89]	2.19	8 1.02
## z0[90]	2.19	8 1.03
## z0[91]	1.05	6 1.00
## z0[92]	1.10	7 1.00
## z0[93]	-1.43	9 1.05
## z0[94]	-1.45	10 1.07
## a0[1]	1.36	90 1.00
## a0[2]	1.45	90 1.00
## a0[3]	1.23	90 1.00
## a0[4]	1.24	88 1.00
## a0[5]	1.60	89 1.00
## a0[6]	1.66	89 1.00
## a0[7]	1.37	89 1.00

## a0[8]	1.37	89 1.00
## a0[9]	1.33	91 1.00
## a0[10]	1.39	88 1.00
## a0[11]	1.33	91 1.00
## a0[12]	1.51	90 1.00
## a0[13]	1.61	91 1.00
## a0[14]	1.60	88 1.00
## a0[15]	1.01	89 1.00
## a0[16]	1.03	91 1.00
## a0[17]	0.42	90 1.00
## a0[18]	0.50	90 1.00
## a0[19]	0.56	90 1.00
## a0[20]	0.54	89 1.00
## a0[21]	1.30	90 1.00
## a0[22]	1.46	90 1.00
## a0[23]	1.25	89 1.00
## a0[24]	1.43	88 1.00
## a0[25]	1.20	90 1.00
## a0[26]	1.22	89 1.00
## a0[27]	0.49	89 1.00
## a0[28]	0.59	91 1.00
## a0[29]	0.55	89 1.00
## a0[30]	0.60	88 1.00
## a0[31]	1.01	90 1.00
## a0[32]	2.19	89 1.00
## a0[33]	0.26	90 1.00
## a0[34]	1.05	89 1.00
## a0[35]	0.36	89 1.00
## a0[36]	1.67	89 1.00
## a0[37]	2.22	89 1.00
## a0[38]	0.99	88 1.00
## a0[39]	1.67	89 1.00
## a0[40]	0.81	89 1.00
## a0[41]	0.99	90 1.00
## a0[42]	0.80	91 1.00
## a0[43]	1.25	91 1.00
## a0[44]	1.31	88 1.00
## a0[45]	0.80	90 1.00
## a0[46]	0.83	90 1.00
## a0[47]	1.58	89 1.00
## a0[48]	1.57	90 1.00
## a0[49]	1.22	89 1.00
## a0[50]	1.21	90 1.00
## a0[51]	1.11	89 1.00
## a0[52]	1.01	89 1.00
## a0[53]	1.06	91 1.00
## a0[54]	1.12	89 1.00
## a0[55]	0.50	90 1.00
## a0[56]	0.41	90 1.00
## a0[57]	1.16	89 1.00
## a0[58]	1.14	91 1.00
## a0[59]	0.97	90 1.00
## a0[60]	1.03	92 1.00
## a0[61]	1.75	90 1.00


```

## a0[62]          1.72    90 1.00
## a0[63]          1.32    90 1.00
## a0[64]          1.36    91 1.00
## a0[65]          2.03    89 1.00
## a0[66]          2.05    89 1.00
## a0[67]          1.52    89 1.00
## a0[68]          1.56    90 1.00
## a0[69]          1.27    90 1.00
## a0[70]          1.21    89 1.00
## a0[71]          1.56    90 1.00
## a0[72]          1.58    90 1.00
## a0[73]          1.40    89 1.00
## a0[74]          1.42    90 1.00
## a0[75]          1.46    89 1.00
## a0[76]          1.44    89 1.00
## a0[77]          1.40    89 1.00
## a0[78]          1.49    90 1.00
## a0[79]          2.22    90 1.00
## a0[80]          2.22    88 1.00
## a0[81]          1.27    90 1.00
## a0[82]          1.29    89 1.00
## a0[83]          1.20    89 1.00
## a0[84]          1.20    92 1.00
## a0[85]          1.56    89 1.00
## a0[86]          1.56    91 1.00
## a0[87]          1.05    88 1.00
## a0[88]          1.07    90 1.00
## a0[89]          2.07    89 1.00
## a0[90]          2.10    90 1.00
## a0[91]          1.59    89 1.00
## a0[92]          1.62    88 1.00
## a0[93]          0.53    90 1.00
## a0[94]          0.49    90 1.00
## lp__            -20424.59   39 1.00
##
## Samples were drawn using NUTS(diag_e) at Thu Apr 27 18:05:31 2023.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

```

Model Comparison

Model 1

```

stan_mlvar_program_MC1<- '
data {
  // number of observations
  int<lower=1> n;
  // response
  vector[n] Sync_standardized;
  // predictor
  vector[n] PS_standardized;
}

```

```

parameters {
  // std of response, single continuous value
  real<lower=0> sigma;
  real b0;
  real b1;

}
model {
  // vector of n values: expected Synchrony for each observation
  vector[n] mu;
  // loop over the n cases in the dataset to estimate mu_i values
  for (i in 1:n) {
    mu[i] = b0 + (b1*PS_standardized[i]);
  }

  // prior for intercept
  b0 ~ normal(0, 0.5);
  // priors for slopes
  b1 ~ normal(0, 1);
  // prior for sigma
  sigma ~ lognormal(0,1);

  // defining likelihood in terms of mus and sigma
  Sync_standardized ~ normal(mu, sigma);
}

generated quantities {
  vector[n] log_lik;
  for (i in 1:n) {
    log_lik[i] = normal_lpdf(Sync_standardized[i] | b0 + b1 *PS_standardized[i], sigma);
  }
}

```

```

mc1_mlvar_model <- stan(model_code = stan_mlvar_program_MC1,
                        data = stan_data,
                        chains = 1,
                        iter = 500,
                        warmup = 250,
                        refresh = 0)

```

```

## Running MCMC with 1 chain...
##
## Chain 1 finished in 11.4 seconds.

```

```

stan_mlvar_program_MC2<- '
data {
  // number of observations
  int<lower=1> n;
  // response
  vector[n] Sync_standardized;
  // predictor

```

```

vector[n] PE_standardized;
// predictor
vector[n] EC_standardized;

}
parameters {
// std of response, single continuous value
real<lower=0> sigma;
real b0;
real b1;
real b2;

}
model {
// vector of n values: expected Synchrony for each observation
vector[n] mu;
// loop over the n cases in the dataset to estimate mu_i values
for (i in 1:n) {
mu[i] = b0 + (b1*PS_standardized[i] + (b2*EC_standardized);
}

// prior for intercept
b0 ~ normal(0, 0.5);
// priors for slopes
b1 ~ normal(0, 1);
b2 ~ normal(0, .5);
// prior for sigma
sigma ~ lognormal(0,1);

// defining likelihood in terms of mus and sigma
Sync_standardized ~ normal(mu, sigma);
}

generated quantities {
vector[n] log_lik;
  for (i in 1:n) {
    log_lik[i] = normal_lpdf(Sync_standardized[i] | b0 + b1 *PS_standardized[i], sigma);
  }
}
}

,

mc2_mlvar_model <- stan(model_code = hier_mlvar_program_MC1,
                        data = stan_data,
                        chains = 1,
                        iter = 500,
                        warmup = 250,
                        refresh = 0)

```

Model 2

```
## Error in serialize(object, connection = NULL, ascii = ascii, version = serializeVersion): object 'hi
```

Comparison

```
compare(mc1_mlvar_model, mc2_mlvar_model, func = WAIC)
```

```
## Error in compare(mc1_mlvar_model, mc2_mlvar_model, func = WAIC): object 'mc2_mlvar_model' not found
```

Interpretation

Based on the WAIC score, the model with the lower WAIC score is generally considered to be the better model as it provides a better balance between the goodness of fit and model complexity. As we can see here, the model that have both standardized eye contact and pupil size is more informative in explaining the response variable.

Present the Posterior

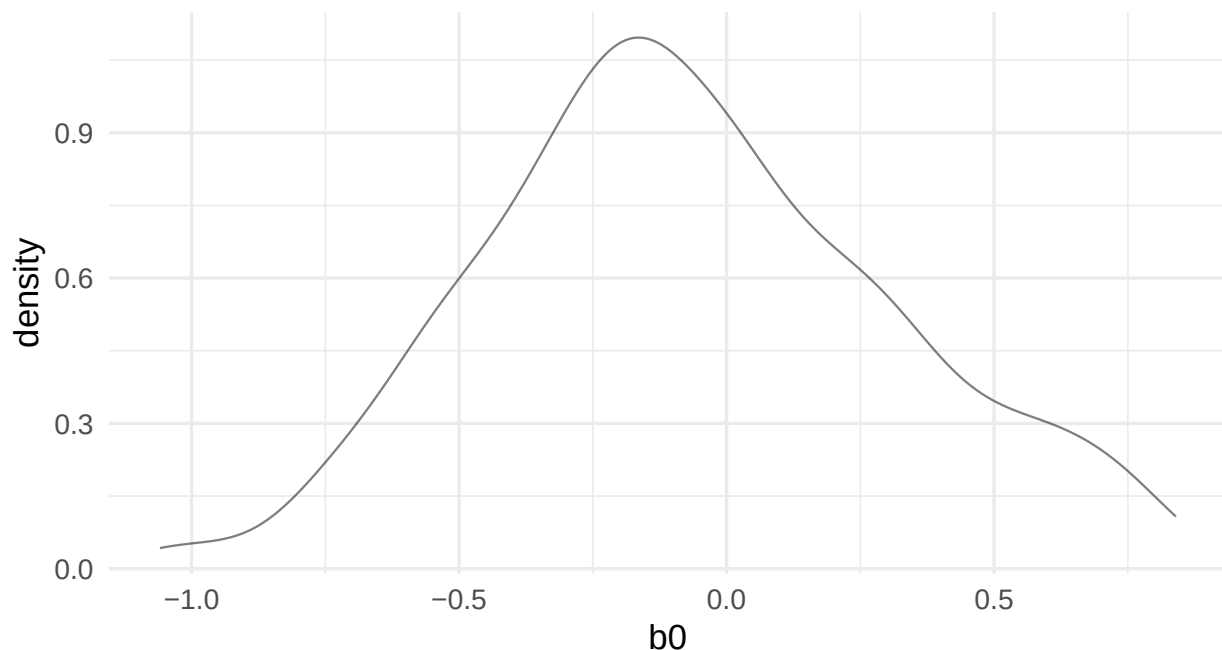
```
psamp <- extract.samples(mlvar_stan_model, n = 10000)
psamp <- data.frame(psamp)
glimpse(psamp)
```

```
## Rows: 500
## Columns: 7
## $ sigma <dbl> 0.983672, 0.982885, 0.983097, 0.984730, 0.980902, 0.987027, 0.98~
## $ b0 <dbl> -0.0788653, -0.1868140, 0.4237970, -0.1169480, -0.3739930, -0.50~
## $ b1 <dbl> 0.0500095, 0.0550878, 0.0565796, 0.0534088, 0.0496411, 0.0571218~
## $ b2 <dbl> 0.166396, 0.174113, 0.161541, 0.166757, 0.178954, 0.172591, 0.16~
## $ b3.1 <dbl> 0.070172500, 0.172571000, -0.435401000, 0.117111000, 0.377585000~
## $ b3.2 <dbl> 0.07884220, 0.18931300, -0.42019700, 0.11725700, 0.37440500, 0.5~
## $ lp__ <dbl> -27276.3, -27277.8, -27278.9, -27275.5, -27279.4, -27277.4, -272~
```

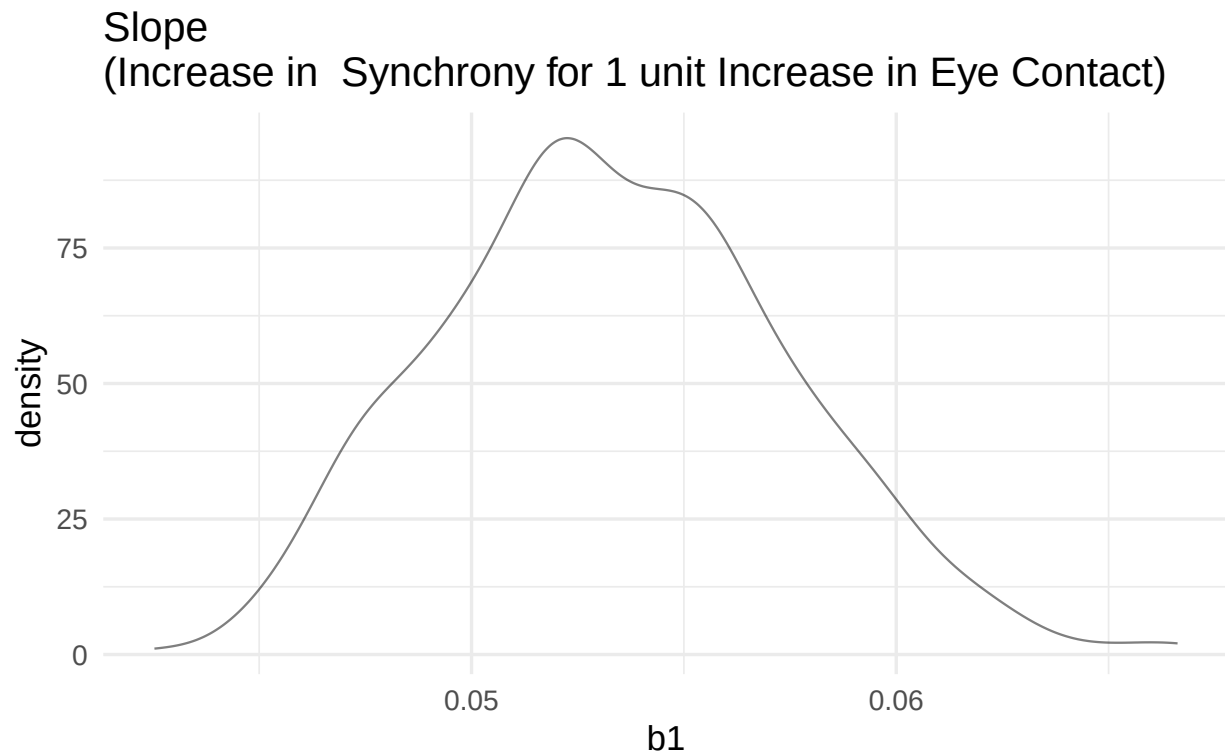
```
gf_dens(~b0, data = psamp, title = 'Intercept\n(Standardize Synchrony with average Pupil Sive and Eye C
```

Intercept

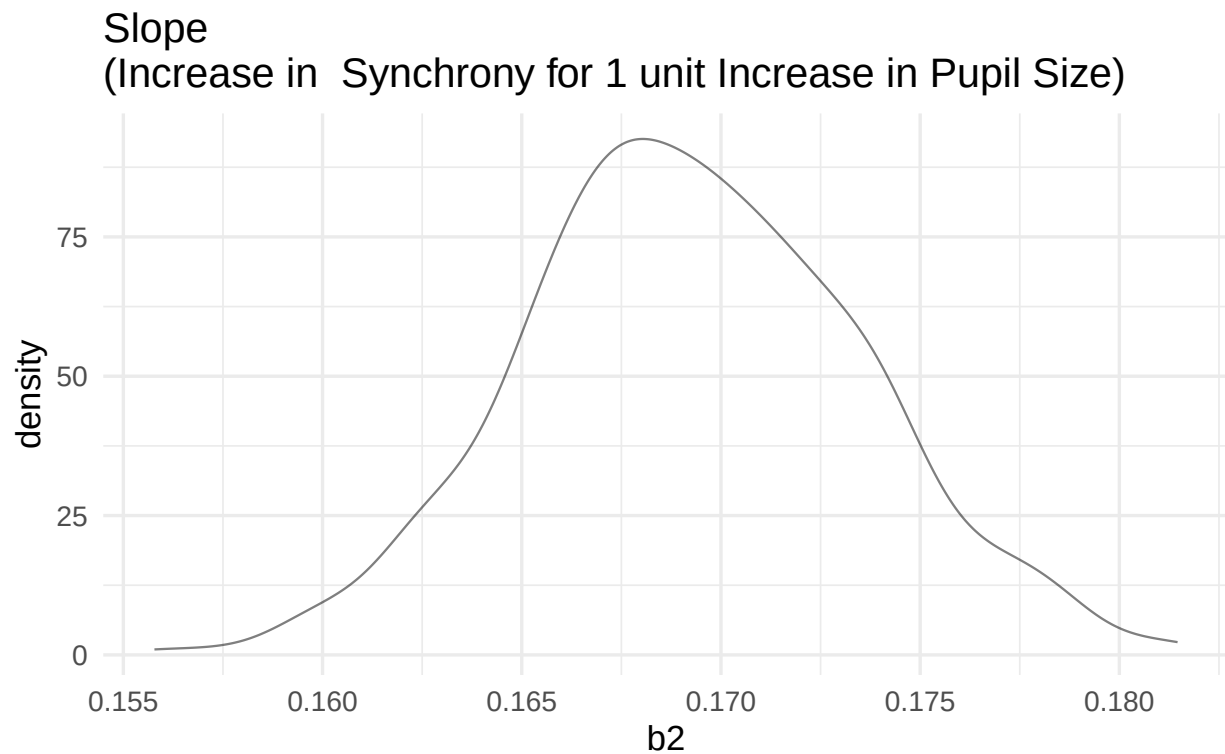
(Standardize Synchrony with average Pupil Sive and Eye Cont



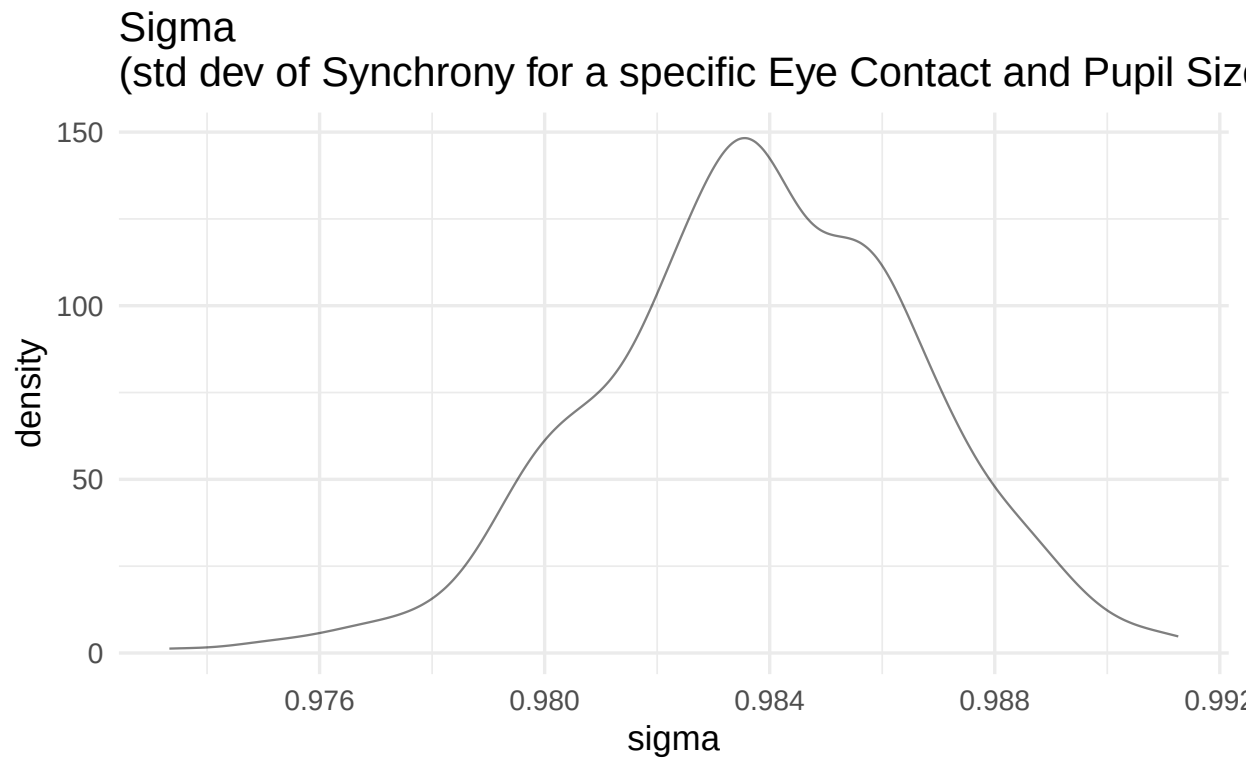
```
gf_dens(~b1, data = psamp, title = 'Slope\n(Increase in Synchrony for 1 unit Increase in Eye Contact)')
```



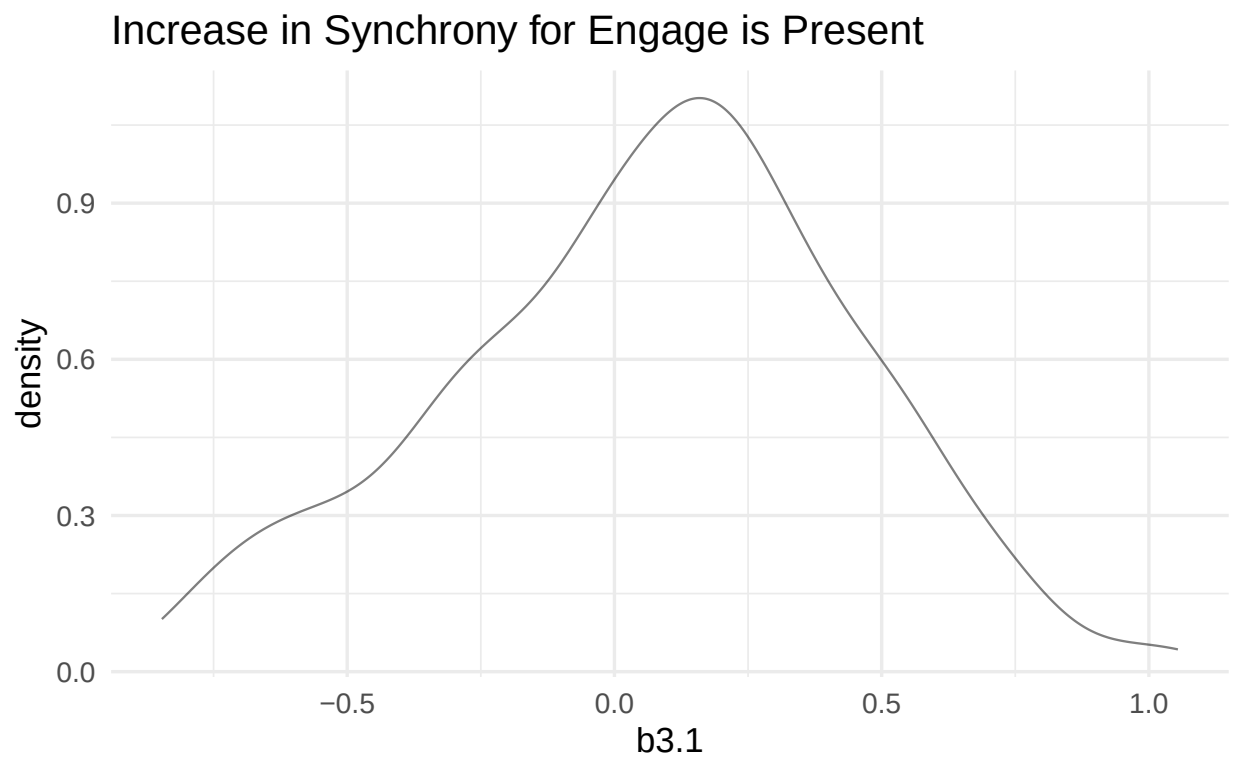
```
gf_dens(~b2, data = psamp, title = 'Slope\n(Increase in Synchrony for 1 unit Increase in Pupil Size)')
```



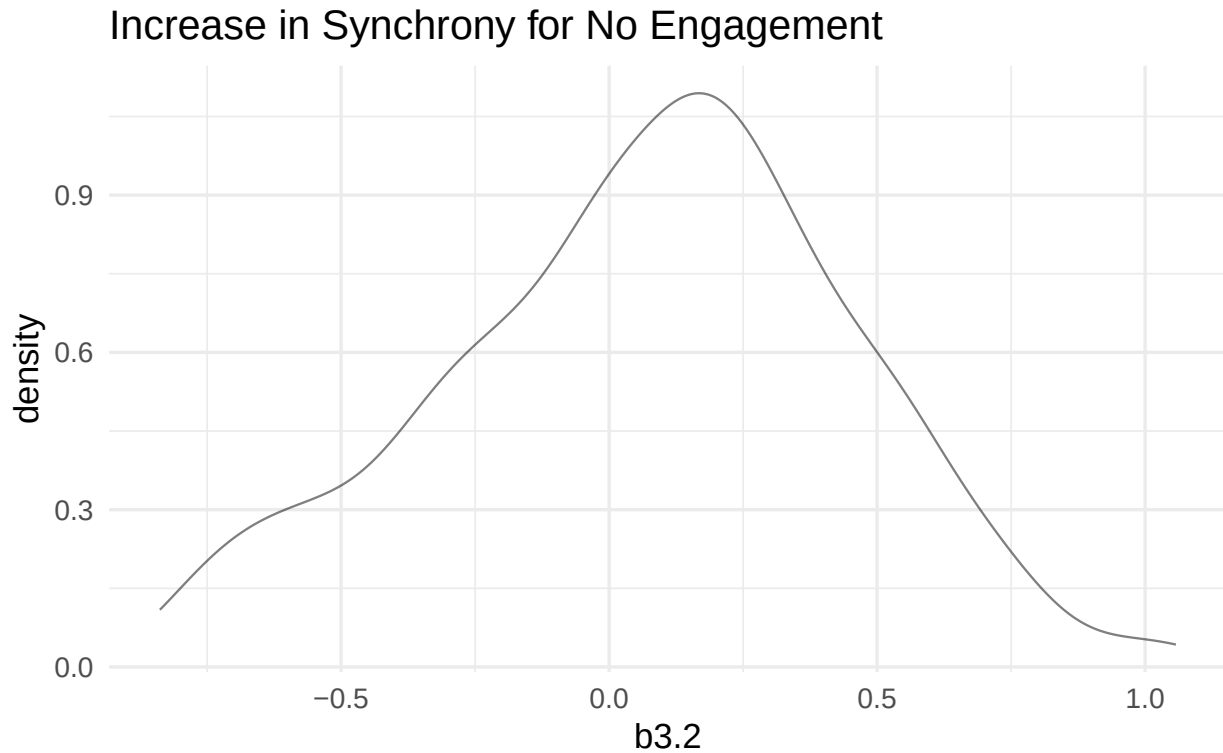
```
gf_dens(~sigma, data = psamp, title = 'Sigma\n(std dev of Synchrony for a specific Eye Contact and Pupil Siz
```



```
gf_dens(~b3.1, data = psamp, title = 'Increase in Synchrony for Engage is Present')
```



```
gf_dens(~b3.2, data = psamp, title = 'Increase in Synchrony for No Engagement')
```



Based on our predictive posterior distribution, an increase in all each of the predictors, increases the number of synchrony.

Discussion or Answer to Research Question

Our Bayesian analysis of the relationship between eye contact and shared attention during conversations suggests that eye contact has a positive effect on the duration of shared attention. Specifically, our posterior distribution for the effect of eye contact suggests synchrony peaks when the eye contact is high.